

Technical Requirement Document: Embodied Drosophila Literacy Simulation (EDLS)

1. Project Vision and Strategic Objective

The convergence of synaptic-resolution connectomics and high-fidelity embodied AI marks a transition from observational neuroscience to the construction of functional, biologically-grounded cognitive agents. While historical AI development has relied on generic, tabula rasa architectures, the EDLS project seeks to exploit the structural inductive biases inherent in the *Drosophila melanogaster* brain to explore the biological substrate of symbolic learning. By grounding a neural controller within a physics-accurate biomechanical agent, we move beyond the triviality of sensorimotor navigation—such as steady-state walking or flight stabilization—into the higher-order cognitive domain of literary pattern recognition and subword assembly. The core mission of the EDLS is to instantiate a functional "digital twin" of the fruit fly that must navigate a 3D browser-based interface to perform linguistic tasks. This requires a fundamental shift from simple reactive behaviors to a system capable of temporal sequence modeling and morphological association. The strategic objective is to determine how a nervous system evolved for spatial navigation and chemical sensing can be computationally repurposed for the assembly of prefixes, roots, and suffixes within a virtual environment. This requirement document defines the anatomical, physical, and algorithmic foundations necessary to realize this embodied literary agent.

2. The Embodied Substrate: Biomechanical and Physics Engine

Biological validity in synthetic intelligence is predicated on the "Digital Sphinx" problem: the phenomenon where high-capacity learning algorithms coerce incompatible neural architectures into producing realistic behaviors through task-overfitting. This was most notably demonstrated when a *C. elegans* connectome was embedded into a *Drosophila* body; while the agent "walked," the solution was biologically meaningless as it bypassed the species-specific sensorimotor routing required for genuine neural emulation. Consequently, high-fidelity physics is the non-negotiable constraint that prevents "unbiological cheats" and ensures the controller respects terrestrial and aerodynamic realities. The EDLS utilizes the **MuJoCo flybody** model, derived from high-resolution confocal fluorescence microscopy of the adult female exoskeleton.

Feature	Specification
Kinematic Complexity	102 Degrees of Freedom (DoFs) Full articulation of head, thorax, 6 legs, and abdominal segments.
Control Action Space	59-Dimensional Torque Vector Decoded efferent commands for specific motor neuron pools.
Joint Arrangement	66 Articulated Joints Includes complex multi-segmented leg joints and wing hinges.
Fluid Dynamics	Low-Reynolds Solvers Phenomenological solvers for instantaneous lift and rotational damping.
Surface Interaction	Adhesion Actuators Simulates van der Waals/capillary forces of distal tarsal pads (pulvilli).

To interact with the virtual smartphone, the agent must generate traction on simulated glass. This is achieved via adhesion actuators that simulate the mechanics of the tarsal pads, providing the friction necessary for precision movement. The "So What?" of this physical rigor is

that realistic mass and viscous drag dictate the energy landscape of the task; the neural controller is forced to adopt efficient motor policies for word assembly, ensuring that learning is a product of a biologically constrained system rather than an algorithmic artifact. The physical body requires a controller capable of interpreting complex sensory streams and executing high-dimensional motor plans.

3. Connectomic Controller: The FlyGM Neural Architecture

To minimize the search space for learning, the EDLS adopts a "structural inductive bias" by instantiating a model based on the adult fruit fly's wiring diagram. The neural controller utilizes the **MaleCNS v1.0** dataset, which consists of **166,691 neurons** and **more than 125 million synapses**. Unlike the FlyWire FAFB, which is limited to the cranial cavity, the MaleCNS includes the entire ventral nerve cord (VNC), providing the 11,700 distinct cell types necessary for whole-body motor modulation and leg joint coordination. The **Fly-connectomic Graph Model (FlyGM)** transforms this static anatomy into a functional simulation by assigning signs to edges based on neurotransmitter profiles. Acetylcholine and Glutamate are mapped to excitatory weights (+), while GABA and Histamine are assigned inhibitory weights (-). The graph is partitioned into three functional tiers:

- **Afferent Neurons:** Processing of 3,335 luminance-sensitive (R1–R6) and 811 color-sensitive (R8) photoreceptor inputs alongside tarsal feedback.
- **Intrinsic Neurons:** Recurrent processing and associative memory across the Mushroom Bodies and Central Complex.
- **Efferent Neurons:** Translation of neural states into the 59-dimensional torque commands required for MuJoCo actuators. The simulation utilizes the **PAM cluster** of dopaminergic neurons as the primary reinforcement mechanism. Correct symbolic associations trigger simulated dopamine transients, gating plasticity across synapses between Kenyon cells and Mushroom Body Output Neurons (MBONs). This biological reward pathway allows the agent to develop persistent memories of subword components. The following section details the environment designed to challenge this neural controller.

4. Interactive Environment: The 3D Language Interface

The virtual interface is optimized for the constraints of **Drosophila vision** rather than human perception. The fly's compound eye consists of **750 ommatidia** arranged in a hexagonal grid, providing a panoramic view but severely limited static resolution. **Optical Configuration**

Requirements:

- **Inter-ommatidial Angle ($\Delta\phi$):** 4.5° .
- **Aperture Size:** $16\ \mu\text{m}$.
- **Focal Length:** $20\ \mu\text{m}$.
- **Microsaccadic Sampling:** The simulation must require the agent to employ active head and body movements to resolve high-frequency graphemes (letters) through temporal contrast changes. The "literary task mechanics" are built upon open-source frameworks: **LibreLingo** for the curriculum hierarchy and **Word Root Workshop** for the morphological assembly of **Prefix + Root + Suffix**. The UI's "So What?" is driven by the geometric convergence of the ommatidial optical axes. This convergence dictates

a receptive field overlap optimized for a **1–10mm distance** , forcing a behavioral trade-off: the fly must choose between "flying to answers" for global navigation or "walking on the screen" for high-acuity symbol resolution. This interaction produces high-resolution telemetry that must be rigorously evaluated.

5. Evaluation Framework: Sequence Modeling and Cognitive Psychometrics

Distinguishing between random behavior and genuine literacy acquisition requires a tiered analytical stack. For high-capacity visual association, we implement the **xLSTM (Extended Long Short-Term Memory)** with **Tiled Flash Linear Attention (TFLA)** kernels to maximize WebGPU efficiency. To ensure numerical stability and prevent overflow during exponential gating, the mLSTM state update must incorporate normalizer (n_t) and stabilizer (m_t) states. **The mLSTM State Update Formula:** $C_t = f_t \odot C_{t-1} + i_t \odot (v_t k_t^T)$
 $n_t = f_t \odot n_{t-1} + i_t$ $m_t = \max(\log f_t + m_{t-1}, \log i_t)$ For adaptive curriculum navigation, we utilize **Decoupled Decision Transformers (DDT)** . To minimize the context window overhead, the DDT will process only the **latest Return-to-Go (RTG)** parameter, which is modulated by PAM cluster dopamine rewards to adjust game difficulty in real-time. Cognitive state is diagnosed using the **G-DINA model** and **Half-Life Regression (HLR)** . Mastery is mapped through a Q-matrix that links behavior to latent attributes. **Table 2: Q-Matrix for Latent Attribute Diagnosis**

Task Description	α_1 : Edge Detection	α_2 : Shape Discrim.	α_3 : Seq. Tracking	α_4 : Motor Targeting
Distinguish 'T' from 'H'	1	1	0	1
Select 'A' then 'B'	1	1	1	1
Touch dark region	1	0	0	1

The **Many-Facet Rasch Model** is required to standardize task difficulty and neural policy capability onto a single objective logit scale, allowing for an absolute measurement of learning progress independent of physics tuning.

6. Web-Based Infrastructure and Telemetry Pipeline

Executing petabyte-scale connectomics in a browser environment requires high-performance hardware acceleration and specialized network protocols.

- **MuJoCo via WebAssembly (WASM):** For client-side, 800Hz physics execution.
- **WebGPU with TFLA:** For real-time xLSTM inference and 3D neural activity reconstruction, maximizing SRAM utilization.
- **SharedArrayBuffer:** For multi-threaded performance in the sensorimotor loop. The telemetry pipeline uses **xAPI** for millisecond-resolution data capture. To prevent packet loss and latency in high-volume transfers, the architecture utilizes a **Science DMZ** . This involves the use of dedicated **Data Transfer Nodes (DTNs)** that bypass **stateful firewall inspections** , routing telemetry directly to the Learning Record Store (LRS).

7. Repository and Framework Manifest

The implementation of the EDLS requires the integration of the following open-source modules: **Connectomics & Physics**

- TuragaLab/flybody: XML-based MuJoCo biomechanical model.
- google-deepmind/mujoco: Native C physics engine core.

- EPFL-Center-for-Imaging/FlyGym: NeuroMechFly v2 for vision and sensory modeling.**Neural Modeling & Learning**
- cobanov/awesome-fly: Repository for connectome-to-graph conversion tools.
- nftechie/doomfly: Connectome-environment interfacing for retinotopic mapping.
- MarkUnthank/flyhard: Connectome-based actuator control for precision movement.**Language Learning & UI**
- LibreLingoCommunity/LibreLingo: Macro-curriculum and exercise architecture.
- joeseesun/word-root-workshop: Morphological assembly logic (Prefix/Root/Suffix).
- SudoKMaar/leaf: Gamification primitives and UI state management.**Evaluation & Infrastructure**
- google-deepmind/acme: Distributed reinforcement learning framework.
- IEEE 9274.1.1: xAPI standard libraries for telemetry streaming.The EDLS architecture demonstrates that biological wiring diagrams are not merely anatomical records but functional blueprints for embodied AI. By grounding symbolic literacy in the *Drosophila* connectome, we establish a scalable roadmap for transforming static biological maps into autonomous, learning agents, providing the definitive operational model for future mammalian research.